CAPSTONE PROJECT · SOFTWARE ENGINEERING

repoCare

Health Insight Platform

An AI-assisted personal health platform that turns medical reports, biomarkers, and wearable sensor data into clear, organized insight.

TEAM

Ameer Alghamdi · 2340533
Abdulamlik Alzahrani · 2340534
Faisal Alorabi · 2341120
Osama Alghamdi · 2340546
Abdullah Alahmadi · 2340221

SUPERVISOR

Dr. Mohammad Alahmadi

INSTITUTION

College of Computer Science & Engineering
University of Jeddah



Medical reports · OCR & biomarkers · Wearable sensors · AI insights · Android

SECTION 01 · PROJECT PLANNING

Health data is everywhere — and nowhere at once.

People collect medical reports from hospitals and labs in many formats, and wearable devices add another stream of data on top. None of it talks to each other, and most of it isn't readable by the person it belongs to.

Fragmented records

Reports live as PDFs and printed sheets across hospitals and labs. No single place to keep them.

Complex to read

Reference ranges, abbreviations, and units force users to Google their own results.

Wearables alone

Heart-rate, SpO_2, and stress data sit isolated in their own apps — disconnected from lab work.

Rising chronic load

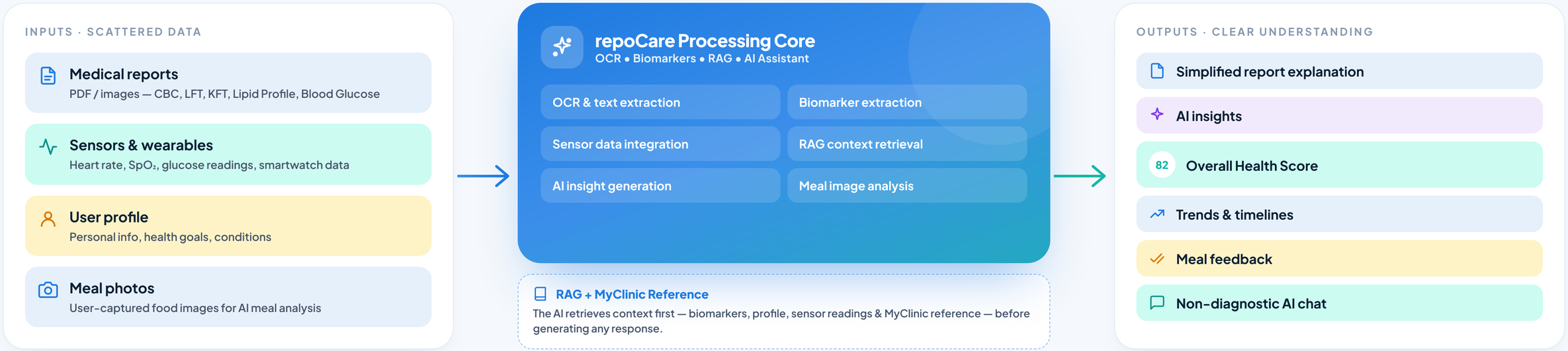
In Saudi Arabia, **≈18.95%** of adults live with at least one chronic disease.

Ministry of Health · Statistical Yearbook 2023

SECTION 01 · PROJECT PLANNING

From scattered health data to clear personal insights.

repoCare organizes medical reports, sensor readings, and user health data into one AI-assisted, non-diagnostic health insight platform.



Informational support only — not diagnosis or treatment.

SECTION 01 · PROJECT PLANNING

What's in — and what's deliberately out.

IN SCOPE

- ✓ Android mobile app
- ✓ PDF & image report upload
- ✓ OCR text extraction & parsing
- ✓ Biomarker classification (Low / Normal / High)
- ✓ Wearable sensors via BLE gateway
- ✓ AI assistant & insights
- ✓ Meal photo analysis
- ✓ Trends, health score & alerts

SUPPORTED TEST TYPES

CBC · LFT · KFT · Lipid Profile · Blood Glucose

OUT OF SCOPE

- ✗ **Medical diagnosis or treatment**
  Explanations are informational, not diagnostic.
- ✗ **Real-time clinical monitoring**
  Not for ICU-grade or continuous vitals.
- ✗ **Hospital-system integration**
  No direct EMR / HIS connectors.
- ✗ **Emergency response**
  No call-for-help or SOS features.
- ✗ **Direct communication with providers**
  No telemedicine or messaging built in.

SECTION 01 · PROJECT PLANNING

Five concrete things we set out to do.

Each objective maps to a deliverable module in the prototype — so the success criteria are measurable, not aspirational.

01

Secure report storage

Mobile platform to upload & centralize medical reports as PDF / image.

02

AI-based interpretation

Analyze reports and produce simplified, non-diagnostic explanations.

03

Sensor integration

Combine wearable physiological data with report-based indicators.

04

AI assistant

Chat that answers questions about a user's own reports clearly.

05

Health awareness

Improve user awareness — without replacing professional medical advice.

Aim → develop an AI-powered health-monitoring platform that lets users upload & manage medical reports, interpret indicators, and track their status by combining report data with sensor inputs.

SECTION 02 · PROBLEM UNDERSTANDING

Who the system serves and touches.

repoCare puts patients at the center, while supporting professionals and labs indirectly through better-organized personal data.

PRIMARY · HIGH INTEREST

Patients & general users

Upload reports, monitor data, interact with the AI assistant. Chronic-disease users (diabetes, hypertension, cardiovascular) benefit most.

Chronic | General wellness

MEDIUM INTEREST

Healthcare professionals

Receive better-organized patient data; the system is a support tool, not a replacement.

LOW INTEREST

Medical labs

Indirect — labs generate the reports the user uploads and interprets.

MEDIUM INTEREST

Health-tech providers

Wearable & sensor makers whose devices integrate into the platform.

HIGH INTEREST

Chronic-disease users

Regular daily tracking of glucose, blood pressure, lipid & kidney indicators.

SECTION 02 · PROBLEM UNDERSTANDING

Where the field is — and where the gap sits.

01

PHR systems are storage, not understanding

Tang et al. — Personal Health Records centralize data, but offer limited analysis or interpretation.

02

AI in healthcare is mostly clinic-facing

Jiang et al. — most AI healthcare work targets hospitals, not everyday personal health monitoring.

03

Wearables don't talk to labs

Pantelopoulos & Bourbakis — wearable physiological data sits separately from medical reports.

04

Saudi market: high need, scattered tools

Sehhaty & NahdiCare are excellent for service access — but not for unified interpretation of personal lab + sensor data.

THE GAP REPOCARE FILLS

A unified, AI-assisted view that no current app puts together.

- ✓ Reports + wearables in one place
- ✓ Plain-language explanation of indicators
- ✓ Continuous monitoring over time
- ✓ AI insights — non-diagnostic, traceable

SECTION 02 · PROBLEM UNDERSTANDING

How repoCare lines up against today's apps.

Five evaluation criteria, drawn directly from common features of digital-health platforms and from our own objectives.

SYSTEM	DATA INTEGRATION	REPORT INTERPRETATION	CONTINUOUS MONITORING	AI INSIGHTS	USER ACCESSIBILITY
Ornament	Limited	Yes	No	Limited	High
Oura	Yes	No	Yes	Limited	High
Sehhaty	Limited	No	Limited	No	High
NahdiCare	Limited	No	Limited	No	High
repoCare	Yes	Yes	Yes	Yes	High

Bottom line — existing apps each cover one slice. repoCare is the only system that unifies reports, sensors, and AI interpretation in a single personal record.

SECTION 02 · PROBLEM UNDERSTANDING

Can we actually build it this semester?

Four lenses say yes: the tools, the workflow, the budget, and the timeline all line up with a capstone-scope prototype.

01

Technical

Mature, well-documented stack: Java & Spring Boot, Firebase auth / Firestore / Storage, Google OCR, BLE sensors via Arduino / Raspberry Pi.

Java · Spring Boot · Firebase · Google OCR

02

Operational

Designed to assist users, not replace clinicians. Workflow is simple: upload → review → ask. Friendly for non-technical users.

03

Economic

Open-source frameworks, free academic tiers, low-cost sensors (MAX30102, Grove GSR), and cloud Firebase — minimal infrastructure spend.

04

Schedule

Scope is a prototype of core features only — fits the academic semester when phased into auth, reports, sensors, and AI.

SECTION 03 · REQUIREMENTS SPECIFICATION

30 functional requirements — grouped by responsibility.

Each group maps to one of the system's modules, so requirements become test-cases against a single owner.

Account & profile — 3 FRs

- **FR1** Secure register with name, email, password.
- **FR2** Login & reach personal dashboard.
- **FR3** Enter age, gender, height, goals, chronic conditions.

Reports & OCR — 3 FRs

- **FR4** Upload PDF / image medical reports.
- **FR5** OCR-extract medical text data.
- **FR6** Auto-identify CBC, LFT, KFT, Lipid, Glucose.

Biomarkers & trends — 5 FRs

- **FR7** Extract biomarker values from reports.
- **FR8** Normalize values to standard units.
- **FR9** Classify as Low / Normal / High.
- **FR10** Persist structured data for analysis.
- **FR11** Visual graphs / timelines over time.

Sensors & gateway — 7 FRs

- **FR12** Integrate heart-rate, oximeter, GSR.
- **FR13** BLE comms sensor ↔ gateway.
- **FR14** Receive HR, SpO_2, stress readings.
- **FR15** Pull glucose via external API.
- **FR16** Clean & normalize sensor readings.
- **FR17** Daily / weekly aggregates.
- **FR18** Unified user health profile store.

AI assistant & insights — 8 FRs

- **FR19** Auto-detect abnormal indicators & alert.
- **FR20** Compute overall health score.
- **FR21** AI analyzes user health profile.
- **FR22** AI chat interface.
- **FR23** Personalized insights & explanations.
- **FR24** Non-diagnostic responses only.
- **FR25** Explain abnormals with retrieved knowledge.
- **FR26** Suggest checkups based on abnormals.

Meal & nutrition — 4 FRs

- **FR27** Upload meal images.
- **FR28** AI image analysis for food & portion.
- **FR29** Generate nutrition & dietary advice.
- **FR30** Rate meal Low / Medium / High risk per profile.

SECTION 03 · REQUIREMENTS SPECIFICATION

Quality attributes that earn user trust.

Numbers, not adjectives — every quality attribute is anchored to a measurable target the system must meet.

Performance

< 2s
UI navigation & sensor updates

< 10s
OCR + parse a 5 MB report

100+
Concurrent users supported

Reliability

99%
Availability during operational hours

3×
Retry on sync failure before notify

0
Data loss on uploaded reports

Security & privacy

- Encrypted health data at rest
- HTTPS for all backend comms
- Authenticated access only
- Each user sees own data only

Usability

< 2 min
First-time account creation

3 taps
From upload to seeing analysis

Labeled axes & units on every chart; clear validation messages.

Scalability

Add new sensors / health APIs without changing core modules.

Plug-in new AI models without service interruption.

Compatibility

Android 10 or later.

Bluetooth Low Energy (BLE).

Screens from 5" to 7".

Maintainability

Modular UI / data / AI separation.

API & component documentation.

Swap AI / OCR without touching UI.

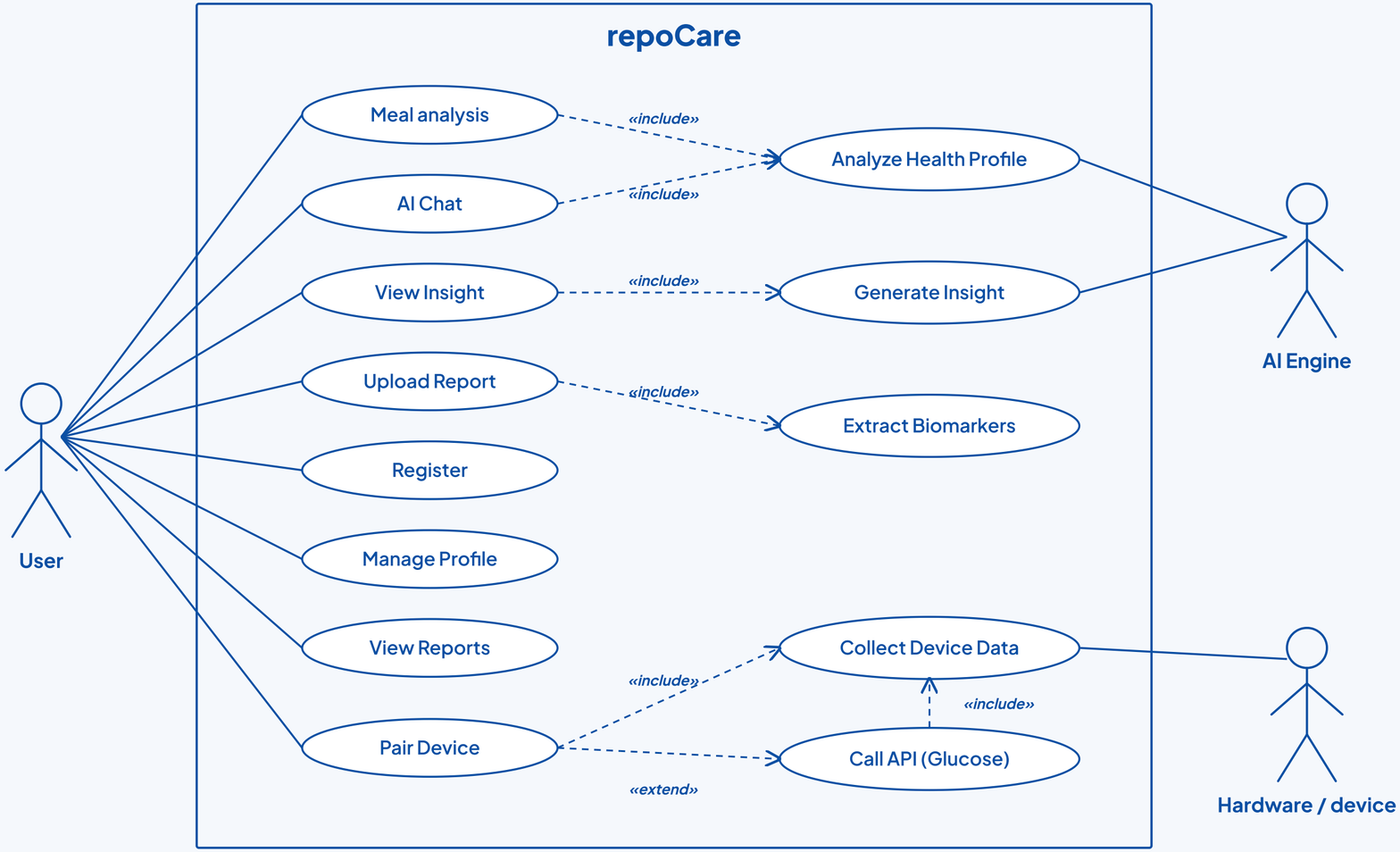
AI INSIGHT LATENCY

< 5s

Insight generation completes within 5 seconds of receiving structured biomarker data — under normal network conditions.

SECTION 04 · UML DIAGRAMS

Who does what with the system.



UML · USE CASE DIAGRAM

Three actors — one human (User), and two system-side (AI Engine, Hardware/device). Use cases group cleanly by responsibility: account & profile, reports, AI work, devices, and insights.

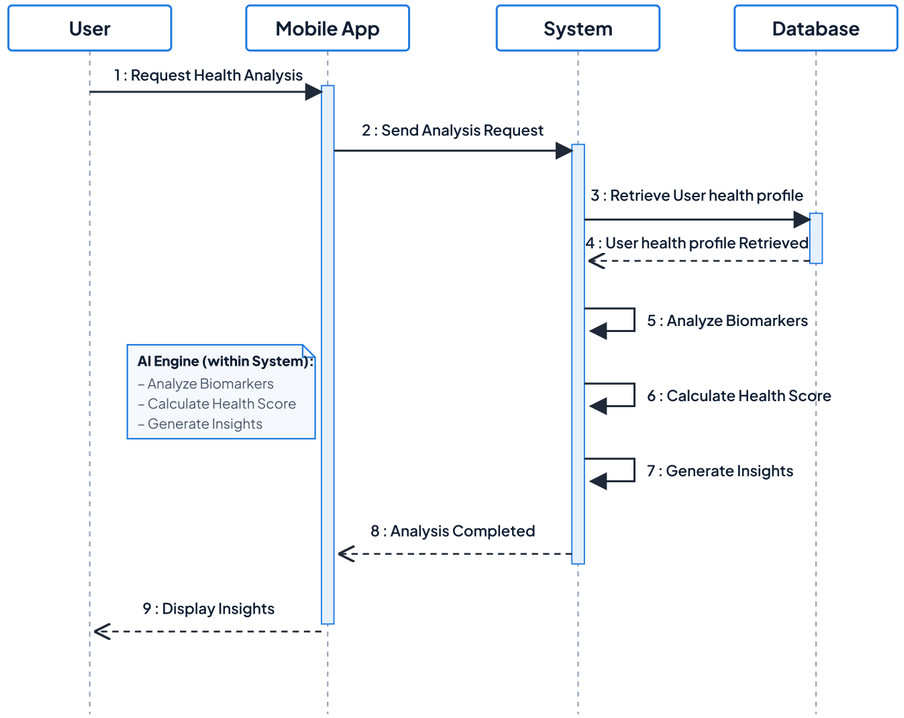
SECTION 04 · UML DIAGRAMS

How parts cooperate — and how reports move through states.

One sequence diagram for the AI health-analysis flow; one state diagram for the medical-report processing lifecycle. Failure is a first-class state.

SEQUENCE DIAGRAM

Health Analysis Process



STATE DIAGRAM

Medical Report Processing

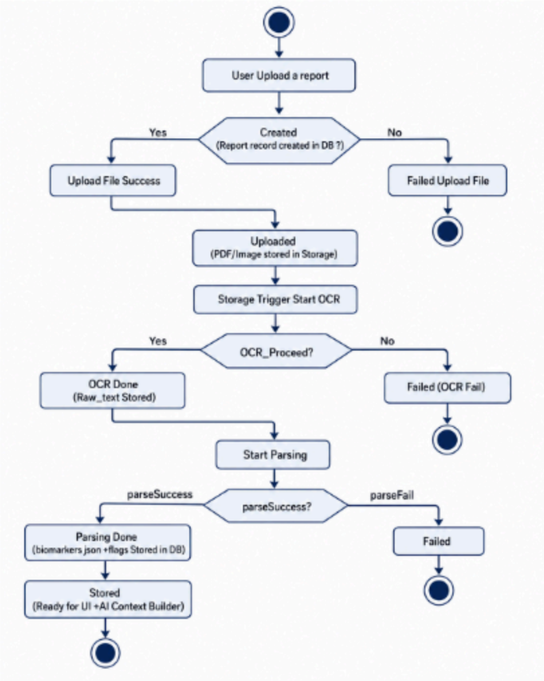


Figure 3.5: Medical Report Processing State

SECTION 04 · UML DIAGRAMS

23 classes, four neighbourhoods.

The static structure of the system, grouped by what each class is responsible for. Edges in the real diagram connect across these neighbourhoods.

User & profile

- `User`
- `UserProfile`

Account info plus age, height, weight, conditions, and health goals.

Reports module

- `MedicalReport`
- `Biomarker`
- `ReportService`
- `ReportProcessor`

Upload → OCR → detect type → extract & normalize biomarkers.

AI module

- `AIEngine`
- `InsightEngine`
- `MealAnalysis`
- `RetrievalManager`
- `KnowledgeBase`
- `ChatSession · Message`

Device module

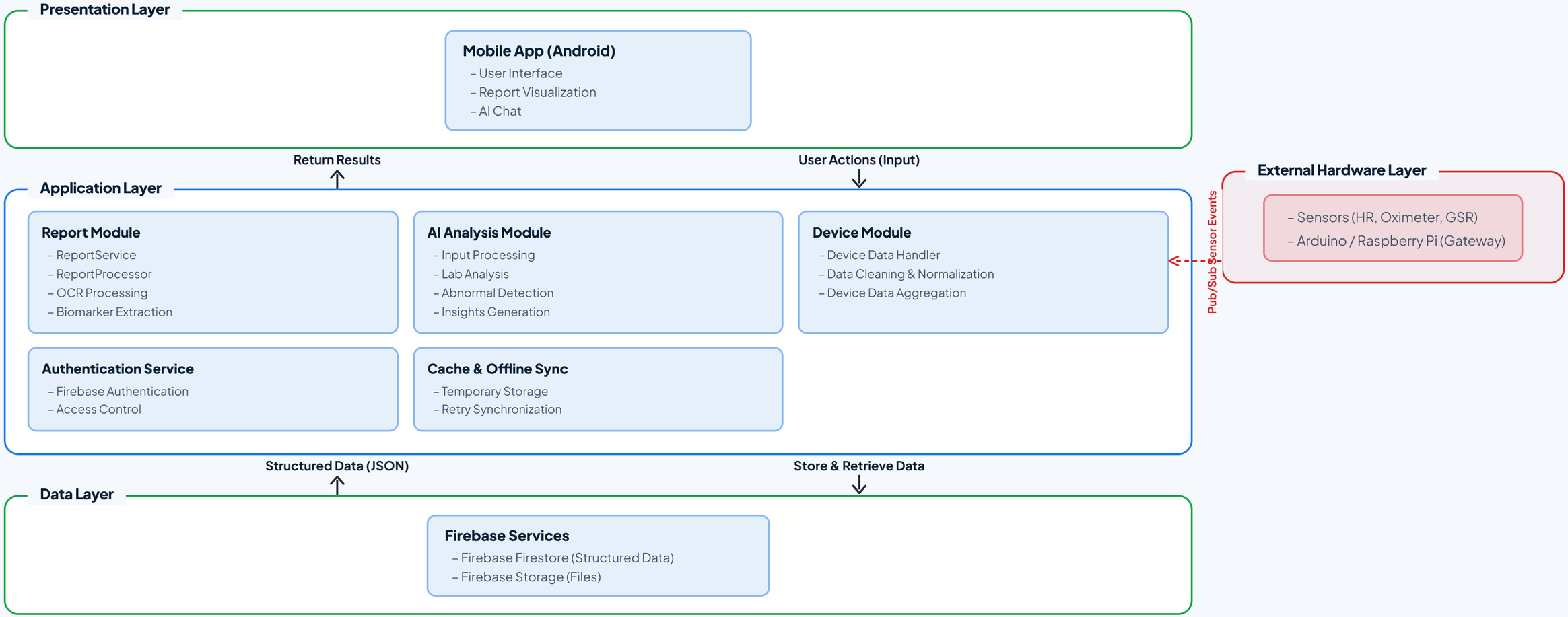
- `Sensor (abstract)`
  - ↳ `MAX30102 · GroveGSR`
- `SensorReading · SensorStatus`
- `BLEManager · Gateway`
- `GatewayAPIClient · GlucoseAPIClient`
- `OfflineBuffer`

Key flows — **Report** upload → process → biomarkers · **Sensor** → Gateway → API client → backend · **AIEngine** pulls profile + reports + sensors + KnowledgeBase to generate insights.

SECTION 05 · DESIGN

A layered core, with event-driven edges.

Four clearly-bounded layers communicate through controlled interfaces. Sensor data uses an asynchronous publisher / subscriber path so the UI never blocks on hardware.



SECTION 05 · DESIGN

Why this hybrid design.

The architecture is intentionally hybrid: a layered pattern keeps the core boring & predictable, while event-driven plumbing handles the messy, real-world parts.

PATTERNS

Layered architecture

Presentation, Application, Data, and External Hardware as 4 clean layers. Adjacent communication only — keeps coupling low.

Event-driven for sensors

Sensors emit measurements asynchronously; the Device Module subscribes and processes them outside the main app flow.

Publisher / Subscriber

A lightweight pub/sub binds sensors to the Device Module — sensors don't know who is listening; the module doesn't block on hardware.

ARCHITECTURAL TACTICS

Modular separation

Each module ships, scales, tests independently.

Auth & access control

Every request authenticated; users see only their own data.

Retry sync

Failed syncs retry up to 3× before user is notified.

Offline buffering

Sensor data caches locally and syncs later when online.

Async event processing

Streaming sensor updates without blocking the UI.

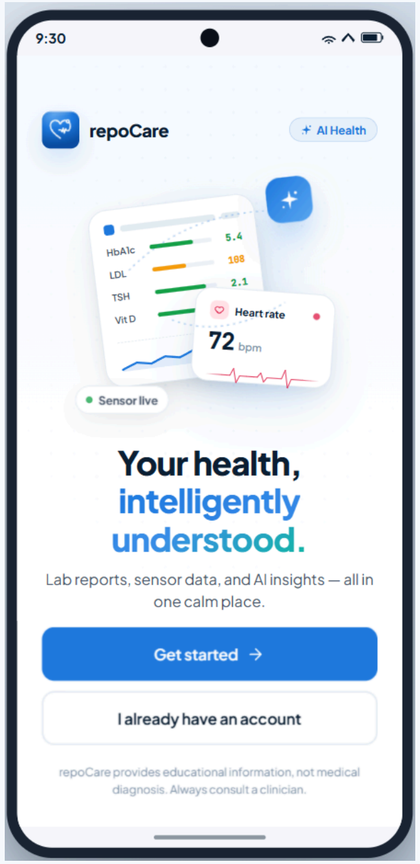
AI safety guardrails

Non-diagnostic prompts; no medical advice generated.

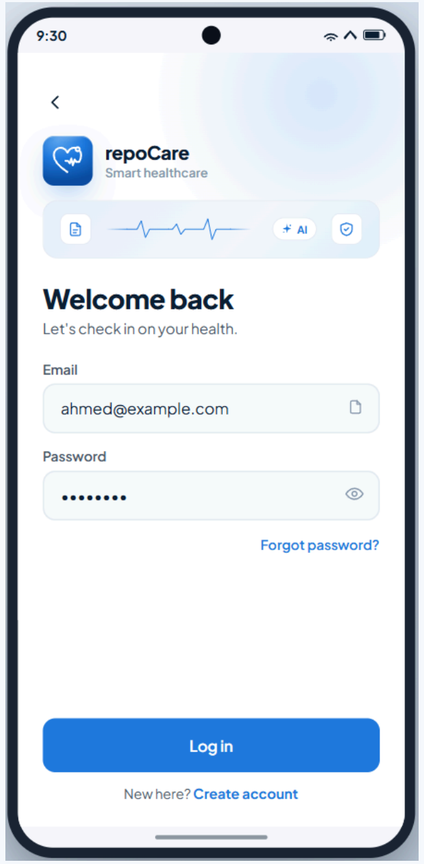
SECTION 06 · GUI PROTOTYPE

A calm entry — onboarding done right.

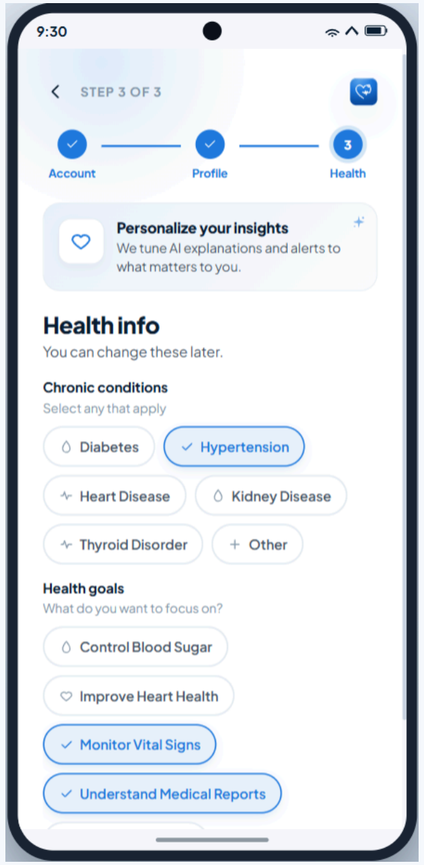
Splash welcomes the user; a familiar login keeps returning users moving; sign-up captures account, profile, then chronic conditions & goals so the AI can tune from day one.



Splash
Welcome · brand · primary CTA



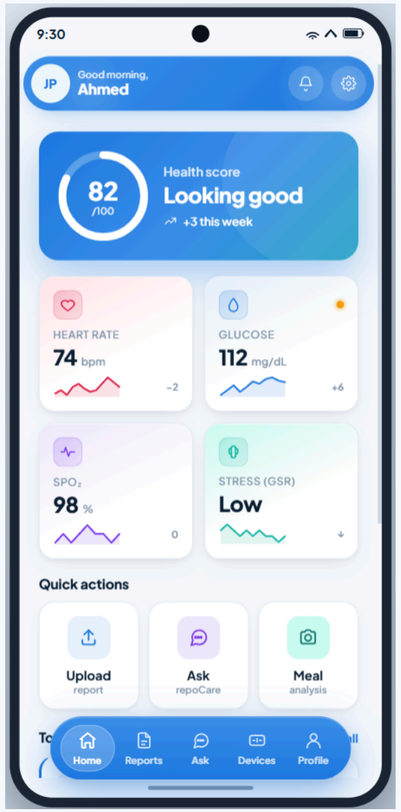
Login
Returning users · forgot password



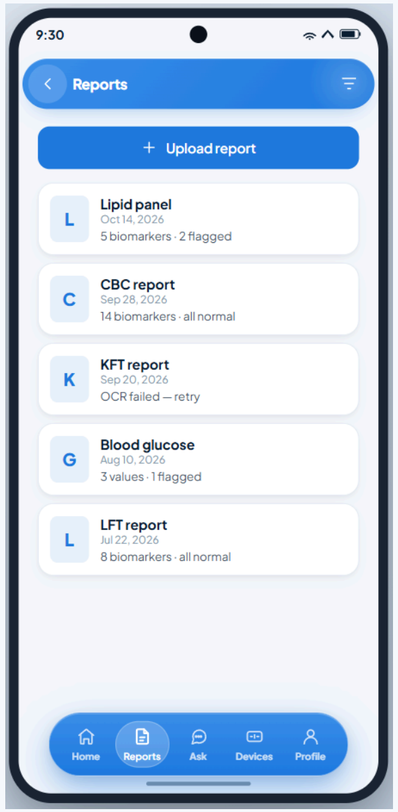
Sign up · Health info
Chronic conditions · health goals

Score, reports, results — in three taps.

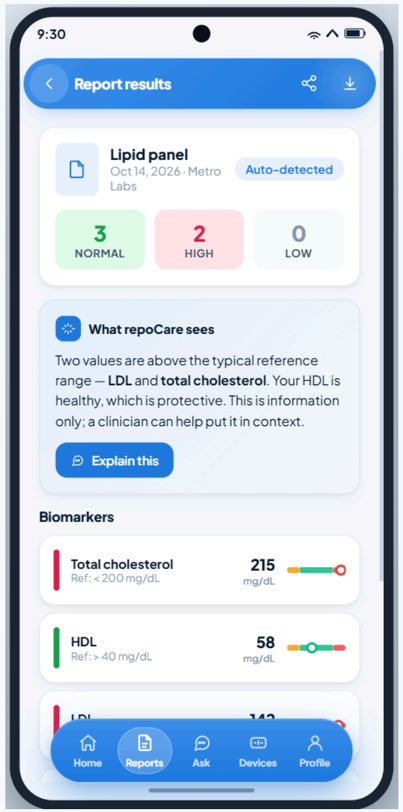
The home dashboard centers the user's health score and live vitals; the reports tab lists every lab; one tap into a report shows grouped biomarkers; one more tap explains a single value in plain language.



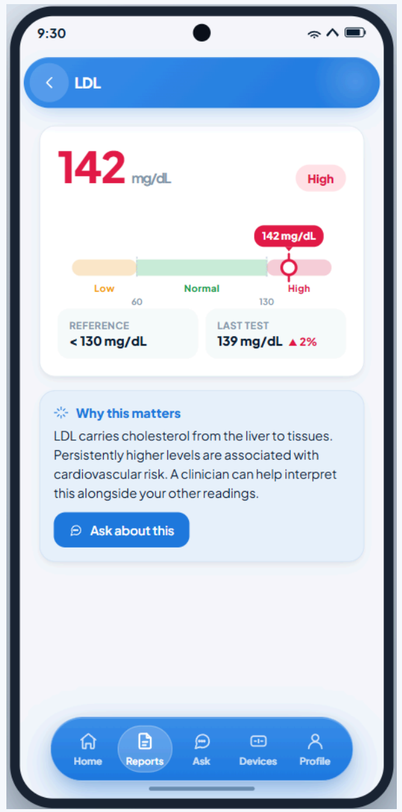
Home
Score · vitals · quick actions



Reports
History · upload · OCR status



Report results
Grouped biomarkers · summary

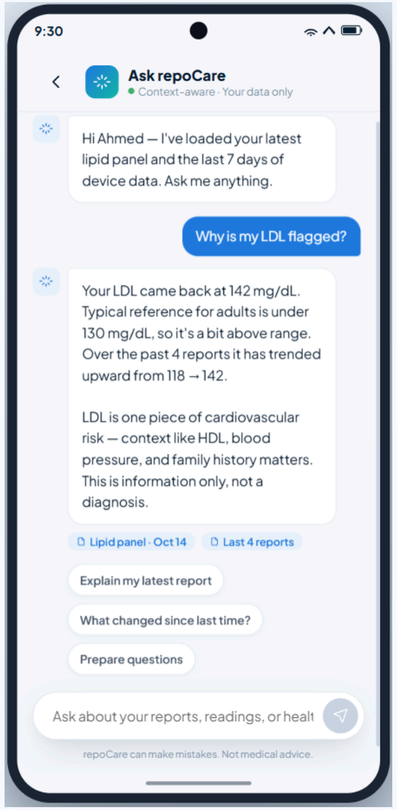


Biomarker detail
Value · range · why it matters

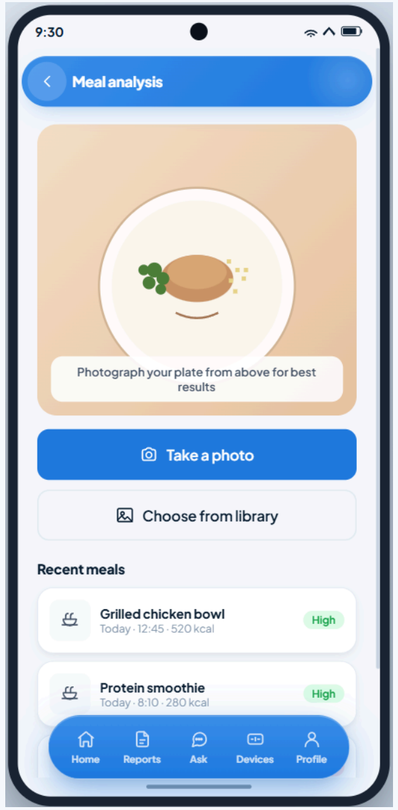
SECTION 06 · GUI PROTOTYPE

repoCare AI chat, grounded in the user's data.

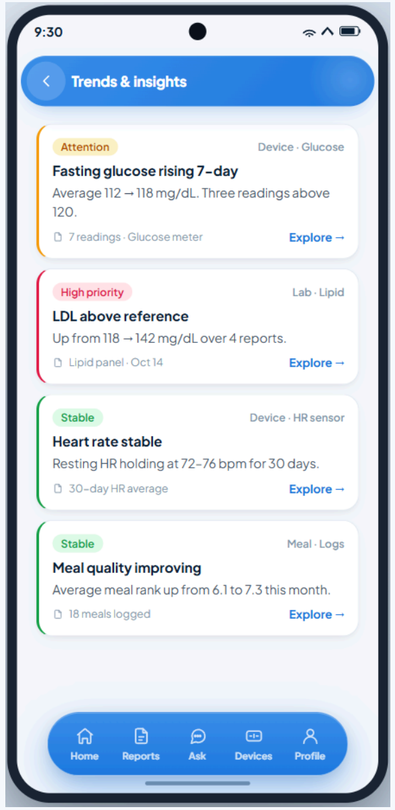
Ask repoCare answers using the user's own reports + recent sensor readings — non-diagnostic, with cited sources. Meal analysis turns a plate photo into nutrition feedback; trends surface what's changing; devices show what's connected and live.



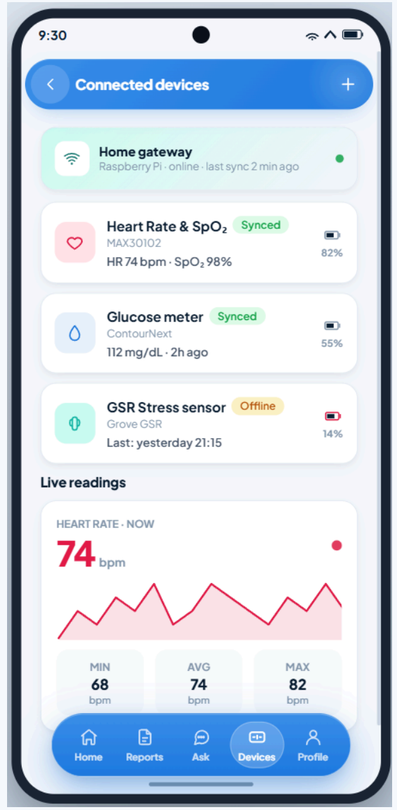
Ask repoCare
Context-aware · cited · non-diagnostic



Meal analysis
Photo → nutrition rank



Trends & insights
Priority · stable · improving



Connected devices
Gateway · sensors · live readings



THANK YOU

Questions?

We'd love to hear your feedback and answer any questions you have about repoCare.

repoCare — Health Insight Platform

College of CSE · University of Jeddah